

DSC-3: PROBABILITY AND STATISTICS

B.Sc. (H) Mathematics | Semester I

Topic-wise Question Bank (2022–2026)

Papers: 2022-23 • 2023-24 • 2024-25 • 2025-26 • Jun 2025

UNIT 1: DESCRIPTIVE STATISTICS

1.1 Stem-and-Leaf Display

[2025–26 | Part 1(a)]

The following table represents the running times (in minutes) of 26 randomly selected recent American movies. Construct a stem-and-leaf plot and comment on interesting features of the display.

Data: 94, 90, 95, 93, 128, 95, 125, 91, 104, 116, 162, 102, 90, 110, 92, 113, 116, 90, 97, 103, 95, 120, 109, 91, 138, 94

[2024–25 | Part 1(a)]

Construct a stem-and-leaf display for the given batch of exam scores, repeating each stem twice. What feature of the data is highlighted by this display?

Data: 74, 89, 80, 93, 64, 67, 72, 70, 66, 85, 89, 81, 81, 71, 74, 182, 85, 63, 72, 81, 81, 95, 84, 81, 80, 70, 69, 66, 60, 83, 85, 98, 84, 68, 90, 82, 69, 72, 87, 88

[2023–24 | Part 1(a)]

The following table gives the accompanying specific gravity values for various wood types used in construction. Construct a stem-and-leaf display and comment on any interesting features of the display.

Data: .31, .35, .36, .36, .37, .38, .40, .40, .40, .41, .41, .42, .42, .42, .42, .42, .43, .44, .45, .46, .46, .47, .48, .48, .48, .51, .54, .54, .55, .58, .62, .66, .66, .67, .68, .75

[2022–23 | Part 1(a)]

A sample of 26 offshore oil workers took part in a simulated escape exercise, resulting in the following data on time (sec) to complete the escape. Construct a stem-and-leaf display and comment on any interesting features.

Data: 389, 356, 359, 363, 375, 424, 325, 394, 402, 373, 373, 370, 364, 366, 364, 325, 339, 393, 392, 369, 374, 359, 356, 403, 334, 397

[Jun 2025 | Part 1(b)]

Do running times of American movies differ from running times of French movies? The author investigated this question by randomly selecting 25 recent movies of each type. Construct a stem-and-leaf plot.

Data: 94, 90, 95, 93, 128, 95, 125, 91, 104, 116, 162, 102, 90, 110, 92, 113, 116, 90, 97, 103, 95, 120, 109, 91, 138

1.2 Measures of Centre and Spread (Median, Quartiles)

[2025–26 | Part 1(b)]

The following data represents the actual blood pressure values for nine randomly selected individuals. Compute the median, upper fourth (Q3), and lower fourth (Q1).

Data: 118.6, 127.4, 138.4, 130.0, 113.7, 122.0, 108.3, 131.5, 133.2

[2024–25 | Part 1(b)]

The following data gives the reasonable award (in \$1000s) in 27 cases identified by the court. What are the values of the quartiles and the fourth spread? How large or small does an observation have to be to qualify as an outlier? As an extreme outlier?

Data: 37, 60, 75, 115, 135, 140, 149, 150, 238, 290, 340, 410, 600, 750, 750, 750, 1050, 1100, 1139, 1150, 1200, 1200, 1250, 1576, 1700, 1825, 2000

[2023–24 | Part 1(b)]

The following data consists of observations on the time until failure (1000s of hours) for a sample of turbochargers. Compute the Median, Upper Fourth (Q3) and Lower Fourth (Q1).

Data: 1.6, 2.0, 2.6, 3.0, 3.9, 3.5, 4.5, 4.6, 4.8, 5.0

[2022–23 | Part 1(b)]

The following data gives the sample of total nitrogen loads (kg N/day) from a particular Chesapeake Bay location. Calculate the median, upper fourth (Q3), and lower fourth (Q1).

Data: 9.69, 13.16, 17.09, 18.12, 23.70, 24.07, 24.29, 26.43, 30.75, 31.54

1.3 Sample Mean, Variance, and Standard Deviation

[2025–26 | Part 1(c)]

Following data consists of values of sodium content (mg) in one serving of cereal for a sample manufactured by a company.

Data: 211, 408, 171, 178, 359, 249, 205, 203, 201

- (i) Calculate sample mean, sample variance, and sample standard deviation.
- (ii) If 50 is added to each observation, compute the new sample variance and standard deviation without reperforming calculations.

[2024–25 | Part 1(c)]

The value of Young's modulus (GPa) was determined for cast plates. Sample observations: 116.6, 115.9, 114.6, 115.2, 115.8.

- (i) Calculate the sample mean, sample variance and sample standard deviation.
- (ii) Subtract 100 from each observation and compute the sample variance of transformed values. Compare with original.

[2023–24 | Part 1(c)]

The following table gives the data on oxidation-induction time (in minutes) for various commercial oils: 87, 103, 130, 160, 180, 195, 132, 145, 211, 105.

- (i) Calculate the sample variance and standard deviation.
- (ii) If the observations were re-expressed in hours, what would be the resulting sample variance and standard deviation? Answer without reperforming calculations.

[2022–23 | Part 1(c)]

Data was collected for the sale of homes in a particular city (in \$1000s): 590, 815, 575, 608, 350, 1285, 408, 540, 555, 679.

- (i) Calculate the sample variance and standard deviation.
- (ii) The evaluator mistakenly forgot to multiply each observation by 5. What would be the resulting values of variance and standard deviation after correction? Answer without reperforming.

[Jun 2025 | Part 1(c)]

Calculate the sample mean, variance and standard deviation from the following data:

203, 130, 160, 180, 195, 132, 145, 211, 105, 145, 153, 152, 138, 87, 99, 93, 119, 129.

[Jun 2025 | Part 1(a)]

Consider a frequency distribution with classes [1,3), [3,5), [5,7), [7,9), [9,11), [11,13), [13,15), [15,17), [17,19) with frequencies 1, 1, 11, 21, 25, 17, 9, 4, 1. Construct an appropriate histogram and comment on any interesting features.

UNIT 2: PROBABILITY

2.1 Basic Probability and Set Operations

[2025–26 | Part 2(a)]

A motorist's route contains two traffic signals. $P(\text{stop at first}) = 0.4$, $P(\text{stop at second}) = 0.5$, $P(\text{stop at at least one}) = 0.7$. Find: (i) $P(\text{stop at both})$, (ii) $P(\text{stop at first but not second})$, (iii) $P(\text{stop at exactly one})$.

[2023–24 | Part 2(a)]

If A and B are any two events, show that $P(A \cap B') = P(A) - P(A \cap B)$. Hence prove that $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

[2024–25 | Part 2(a)]

A certain system can experience three types of defects. $P(A_1) = .12$, $P(A_2) = .07$, $P(A_3) = .05$, $P(A_1 \cup A_2) = .13$, $P(A_1 \cup A_3) = .14$, $P(A_2 \cup A_3) = .10$, $P(A_1 \cap A_2 \cap A_3) = .01$.

- (i) What is the probability the system has both type-1 and type-2 defects but not type-3?
- (ii) What is the probability the system has at most 2 of these defects?

[2022–23 | Part 2(a)]

A consulting firm has bids on 3 projects. $P(A_1) = .22$, $P(A_2) = .25$, $P(A_3) = .28$, $P(A_1 \cap A_2) = .11$, $P(A_1 \cap A_3) = .05$, $P(A_2 \cap A_3) = .07$, $P(A_1 \cap A_2 \cap A_3) = .01$. Compute $P(A_1' \cap A_2')$ and $P(A_1 \cup A_2 \cup A_3)$.

[Jun 2025 | Part 2(a)]

Find the sample space for rolling of two dice and tossing of three coins.

[Jun 2025 | Part 2(b)]

Suppose 55% of adults regularly consume coffee, 45% regularly consume carbonated soda, and 70% regularly consume at least one of these products. (i) What is the probability that a randomly selected adult regularly consumes both? (ii) What is the probability that a randomly selected adult doesn't regularly consume at least one of these products?

[Jun 2025 | Part 2(c)]

Given $P(A) = 0.6$, $P(B) = 0.4$, $P(A \cap B) = 0.3$, $P(C) = 0.2$, $P(A \cap C) = 0.15$, $P(B \cap C) = 0.1$, $P(A \cap B \cap C) = 0.08$. Find $P(A|B)$, $A(B|A)$, $P(A|B \cup C)$.

2.2 Conditional Probability and Independence

[2025–26 | Part 2(c)]

An oil exploration company has two active projects (Asia and Europe). A and B are independent events with $P(A) = 0.4$, $P(B) = 0.7$.

- (i) What is the probability that at least one project will be successful?
- (ii) Given at least one is successful, what is the probability that only the Asian project is successful?

[2024–25 | Part 2(c)]

If A and B are independent events, show that: (i) A' and B are also independent, (ii) A and B' are also independent, (iii) A' and B' are also independent.

[2022–23 | Part 2(c)]

Components are shipped in batches of ten. 50% of batches have no defective components, 30% have one defective, and 20% have two defective. Two components are randomly selected and tested. Find probabilities that the batch has 0, 1, and 2 defective components given one of the two tested is defective.

2.3 Bayes' Theorem

[2025–26 | Part 2(b)]

State Bayes' Theorem. A reviews editor decides whether a review should be short (1-2 pages), medium (3-4 pages), or long (5-6 pages). 60% are short, 30% medium, 10% long. For short reviews, 80% are in Word; for medium, 50%; for long, 30%.

- (i) What is the probability a selected review was submitted in Word format?
- (ii) Given it was in Word format, what is the posterior probability of it being short?

[2024–25 | Part 2(b)]

State Bayes' Theorem. At a gas station, 40% use regular (A_1), 35% use plus (A_2), 25% use premium (A_3). Of regular customers, 30% fill their tanks; of plus, 60%; of premium, 50%. If the next customer fills the tank, what is the probability that regular gas is requested?

[2023–24 | Part 2(b)]

70% of light aircraft that disappear while in flight are subsequently discovered. Of discovered aircraft, 60% have an emergency locator, while 90% of undiscovered aircraft do not. If a light aircraft has disappeared and has an emergency locator, what is the probability it will not be discovered?

[2023–24 | Part 2(c)]

State Bayes' Theorem. A large timeshare operator: 20% of purchasers select a day visit, 50% one-night, 30% two-night. 10% of day visitors, 30% of one-night, and 20% of two-night ultimately purchase. Given a visitor made a purchase, how likely is it that the person made a day visit?

[2022–23 | Part 2(b)]

State Bayes' Theorem. An individual has three email accounts. 70% of messages come into A1, 20% into A2, 10% into A3. Spam rates: A1 = 1%, A2 = 2%, A3 = 5%. A random mail is found to be spam. What is the probability it came in account A2?

UNIT 3: DISCRETE RANDOM VARIABLES & DISTRIBUTIONS

3.1 PMF, CDF, and Expectation

[2025–26 | Part 3(a)]

A certain brand of upright freezer is available in three capacities: 450L, 500L, 550L. X = rated capacity with $P(450) = 0.2$, $P(500) = 0.5$, $P(550) = 0.3$. If price = $2.5X - 650$, what is the expected price and variance of the price paid by the next customer?

[2024–25 | Part 3(b)]

A certain brand of upright freezer: $p(450) = .2$, $p(500) = .5$, $p(550) = .3$. (i) Compute $E(X)$, $E(X^2)$ and $V(X)$. (ii) If price = $2.5X - 650$, what is the expected price paid by the next customer?

[2023–24 | Part 3(b)]

The n candidates for a job have been ranked 1, 2, 3, ..., n . Each has an equal chance of being selected. X = rank of selected candidate. (i) Find the pmf of X . (ii) Compute $E(X)$ and $V(X)$.

[2022–23 | Part 3(c)]

For any random variable X , let $E(X) = 5$ and $E[X(X - 1)] = 27.5$. Compute: (i) $E(X^2)$, (ii) $V(X)$, (iii) $V(2X + 3)$.

[2022–23 | Part 3(b)]

$X = 1$ if a randomly selected vehicle passes an emission test, $X = 0$ otherwise. $p(1) = p$, $p(0) = 1 - p$. (i) Name the random variable. (ii) Compute $E(X^2)$. (iii) Show that $V(X) = p(1 - p)$.

3.2 Geometric Distribution

[2024–25 | Part 3(a)]

Starting at a fixed time, the gender of each new-born child at a certain hospital is observed until a boy is born. Assume successive births are independent. X = number of births observed. Find: (i) the pmf of X , (ii) the cdf of X .

[2022–23 | Part 3(a)]

Starting at a fixed time, observe the gender of each new born child at a hospital until a boy (B) is born. Let $p = P(\text{boy})$. X = number of births up to and including that of the first boy. (i) Find the pmf of X . (ii) Determine the cdf of X .

[Jun 2025 | Part 3(c)]

Starting at a fixed time, the gender of each newborn child is observed at a certain hospital until a boy (B) is born. Let $p = P(\text{boy is born}) = P(B)$. X = number of births up to and including that of the first boy. (i) Find the pmf of X . (ii) Determine the cdf of X .

3.3 Binomial Distribution

[2025–26 | Part 3(b)]

Twenty percent of all telephones of a certain type are submitted for service while under warranty. Of these, 60% can be repaired, 40% must be replaced. If a company purchases ten of these telephones, what is the probability that exactly two will end up being replaced under warranty?

[2023–24 | Part 5(c)]

If X is a binomially distributed random variable with parameters n and p , prove that: (i) $E[X] = np$, (ii) $V[X] = np(1 - p)$.

[2023–24 | Part 5(a)]

If 75% of all purchases in a certain store are made with a credit card and X = number among ten randomly selected purchases made with a credit card is a Binomial variate, determine: (i) $E(X)$, (ii) $V(X)$, (iii) σ_x , (iv) $P(X$ within 1 standard deviation of mean).

[Jun 2025 | Part 4(a)]

Compute the following binomial probabilities directly from the formula for $b(x; n, p)$: (i) $b(3; 6, 0.5)$, (ii) $P(3 \leq X)$ when $n = 6$ and $p = 0.5$, (iii) $P(X \leq 1)$ when $n = 6$ and $p = 0.5$.

3.4 Poisson Distribution

[2025–26 | Part 3(c)]

Let X be a Poisson random variable with parameter λ . Find the mean and variance of X .

[2024–25 | Part 3(c)]

Suppose the number of drivers travelling between a particular origin and destination has a Poisson distribution with parameter $\mu = 20$. What is the probability that the number of drivers will be: (i) between 10 and 20, both inclusive? (ii) within 2 standard deviations of the mean?

[Jun 2025 | Part 3(a)]

Define Poisson distribution. Find the mean and variance of a Poisson distributed random variable X . Are they equal?

3.5 Hypergeometric and Other Discrete Distributions

[2023–24 | Part 3(a)]

In a group of five potential blood donors a, b, c, d, e , only a and b have $O+$ blood type. Five blood samples will be typed in random order until an $O+$ individual is identified. Y = number of typings necessary. (i) Find the pmf of Y . (ii) Draw the line graph and probability histogram of the pmf.

[2022–23 | Part 5(b)]

An insurance company: X = number of months between successive payments. CDF given as $F(x) = 0$ for $x < 1$; 0.30 for $1 \leq x < 3$; 0.40 for $3 \leq x < 4$; 0.45 for $4 \leq x < 6$; 0.60 for $6 \leq x < 12$; 1 for $x \geq 12$. (i) Determine the pmf of X . (ii) Using only the cdf, compute $P(3 \leq X \leq 6)$.

UNIT 4: CONTINUOUS RANDOM VARIABLES & DISTRIBUTIONS

4.1 PDF, CDF, and Expectation

[2025–26 | Part 4(a)]

The current X in a certain circuit has pdf: $f(x) = 0.075x + 0.2, 3 \leq x \leq 5$; 0 otherwise.

- (i) Verify that the total area under the density curve is 1.
- (ii) Calculate $P(X \leq 4)$ and $P(3.5 \leq X \leq 4.5)$.

[2025–26 | Part 4(b)]

X = time a book on two-hour reserve is checked out. CDF: $F(x) = 0$ for $x < 0$; $x^2/4$ for $0 \leq x < 2$; 1 for $x \geq 2$.

(i) Calculate $P(X > 1.5)$. (ii) Obtain the pdf $f(x)$. (iii) Calculate $E(X)$ and $V(X)$.

[2022–23 | Part 4(a)]

The error involved in making a certain measurement is a continuous RV with pdf: $f(x) = 0.09375(4 - x^2)$, $-2 \leq x \leq 2$; 0 otherwise.

(i) Obtain the CDF $F(x)$. (ii) Compute $P(-1 < X < 1)$. (iii) Compute $E[X]$. (iv) Compute $V[X]$.

[2024–25 | Part 4(a)]

'Time headway' in traffic flow: $f(x) = K/x^4$ for $x > 1$; 0 for $x \leq 1$.

(i) Determine the value of K for which $f(x)$ is a pdf. (ii) Obtain the cdf. (iii) Determine $P(X > 2)$ and $P(2 < X < 3)$.

[2023–24 | Part 4(a)]

The distribution of gravel (in tons) sold by a construction supply company in a given week: $f(x) = (3/2)(1 - x^2)$, $0 \leq x \leq 1$; 0 otherwise. Find: (i) CDF, (ii) $E(X)$, (iii) $V(X)$, (iv) σ_x .

[Jun 2025 | Part 4(b)]

Find k such that $f(x) = kx^2$, $0 \leq x \leq 2$; 0 elsewhere, is a probability density function. Also determine $P(0 < x < 4)$.

[Jun 2025 | Part 5(a)]

'Time headway' for two consecutive cars on a freeway: $f(x) = 0.15e^{-0.15(x - 0.5)}$ for $x \geq 0.5$; 0 otherwise. Find the probability that headway time is at most 5 sec.

4.2 Uniform Distribution

[Jun 2025 | Part 5(b)]

Let X , the thickness of a certain metal sheet, have a uniform distribution on $[A, B]$. Write the pdf of X and find the CDF of X .

4.3 Exponential Distribution

[2024–25 | Part 4(c)]

Suppose component lifetime is exponentially distributed with parameter λ . After putting the component into service, leave for a period of t_0 hours and then return to find the component still working. What is the probability that it lasts at least an additional t hours?

[2024–25 | Part 5(b)]

The stress range in a railway bridge connection is exponentially distributed with an average of 6 MPa. Find the probability that: (i) stress range is at most 10 MPa, (ii) stress range is between 5 and 10 MPa.

[2022–23 | Part 4(c)]

Define exponential distribution. Find the mean and standard deviation of an exponentially distributed random variable X . Are they equal?

[Jun 2025 | Part 4(c)]

Define exponential distribution. Find the mean and standard deviation of an exponential distributed random variable X . Are they equal?

4.4 Normal Distribution

[2025–26 | Part 4(c)]

In a road-paving process, X = truck haul time is normally distributed with mean 8.46 min and standard deviation 0.913 min. (i) What is the probability that haul time will be at least 10 min? (ii) What is the probability that haul time will exceed 15 min?

[2025–26 | Part 5(c)]

The probability that a patient recovers from a rare blood disease is 0.4. If 100 people are known to have contracted this disease, use the normal approximation to the binomial to determine the probability that less than 30 survive.

[2025–26 | Part 5(a)]

It has been reported that the Lognormal distribution best describes the distribution of maximum pit depth (mm) of buried pipelines with $\mu = 0.353$ and $\sigma = 0.754$. (i) What are the mean value and variance of pit depth? (ii) What value c is such that only 1% of all specimens have a maximum pit depth exceeding c ?

[2025–26 | Part 5(b)]

Given a standard normal distribution, find the area under the z -curve which lies: (i) between $z = -1.50$ and $z = 2.00$, (ii) to the right of $z = 1.75$, (iii) to the left of $z = -1.39$.

[2024–25 | Part 4(b)]

X = truck haul time, normally distributed with mean 8.46 min and standard deviation 0.913 min. What is the probability that haul time: (i) is at least 10 min? (ii) exceeds 15 min? (iii) remains between 8 and 10 min?

[2024–25 | Part 5(c)]

X = temperature in $^{\circ}\text{C}$; Y = temperature in $^{\circ}\text{F}$. Conversion: $Y = 1.8X + 32$. (i) If median of Y distribution is 50, find median of X distribution. (ii) If third quartile for X distribution is 15, find the third quartile for Y distribution.

[2023–24 | Part 4(b)]

The reaction time for an in-traffic response to a brake signal can be modelled with a normal distribution with mean 1.25 sec and standard deviation 0.46 sec. What is the probability that reaction time is between 1.00 sec and 1.75 sec? If 2 sec is a critical long reaction time, what is the probability actual reaction time will exceed this value?

[2022–23 | Part 5(c)]

The weight distribution of parcels sent is normal with mean 12 lb and standard deviation 3.5 lb. The parcel service wishes to establish a weight value c beyond which there will be a surcharge. What value of c ensures 99% of all parcels are under the surcharge weight?

[2023–24 | Part 5(c)]

The amount of distilled water dispensed by a certain machine is normally distributed with mean 64 oz and standard deviation 0.78 oz. What container size c will ensure overflow occurs only 0.5% of the time?

[Jun 2025 | Part 5(c)]

The amount of distilled water dispensed by a certain machine is normally distributed with mean value 64 oz and standard deviation 0.78 oz. What container size c will ensure that overflow occurs only 0.5% of the time?

[2022–23 | Part 4(b)]

Suppose 25% of all students at a large public university receive financial aid. For a random sample of size 50, use normal approximation to binomial to find: (i) $P(\text{at most 10 students receive aid})$, (ii) $P(\text{between 5 and 15, both inclusive, receive aid})$.

[Jun 2025 | Part 6(a)]

The distribution of egg weights (g) is normal with mean 53 and standard deviation 0.3. $X_1, X_2, \dots, X_{12}$ denote weights of a dozen randomly selected eggs. Find the mean and standard deviation of the sample mean weight $\bar{X}$. Also find the probability that the sample mean weight exceeds 53.5 g.

UNIT 5: SAMPLING DISTRIBUTIONS AND THE CENTRAL LIMIT THEOREM

5.1 Central Limit Theorem

[2025–26 | Part 6(a)]

State the Central Limit Theorem. A soft-drink vending machine dispenses X with mean 200 ml and standard deviation 15 ml. What is the probability that the mean amount dispensed $\bar{X}$ in a random sample of size 36 is at least 204 ml?

5.2 Sampling Distribution of the Mean

[2023–24 | Part 6(c)]

The inside diameter of a randomly selected piston ring has mean 12 cm and standard deviation 0.04 cm. $\bar{X}$ is the sample mean for $n = 16$ rings. (i) Where is the sampling distribution of $\bar{X}$ centered? (ii) What is the standard deviation of the $\bar{X}$ distribution? (iii) How likely is it that the sample mean diameter exceeds 12.01?

[2022–23 | Part 6(c)]

The amount of a particular impurity in a batch of chemical product has mean 4.0 g and standard deviation 1.5 g. $\bar{X}$ = sample mean impurity for a random sample of 50 batches. (i) Where is the sampling distribution of $\bar{X}$ centered? (ii) What is the standard deviation of $\bar{X}$? (iii) What is the probability that $\bar{X}$ is between 3.5 and 3.8 g?

[2024–25 | Part 6(a)]

The lifetime of a certain type of battery is normally distributed with mean 10 hours and standard deviation 1 hour. There are four batteries in a package. What lifetime value is such that the total lifetime of all batteries in a package exceeds that value for only 5 percent of all packages?

UNIT 6: REGRESSION AND CORRELATION

6.1 Correlation Coefficient

[2025–26 | Part 6(c)]

Determine whether there is a relationship between time (minutes) to complete a form in the morning (x) and in the late afternoon (y). Compute and interpret the sample correlation coefficient.

x : 8.2, 9.6, 7.0, 9.4, 10.9, 7.1, 9.0, 6.6, 8.4, 10.5

y : 8.7, 9.6, 6.9, 8.5, 11.3, 7.6, 9.2, 6.3, 8.4, 12.3

[2024–25 | Part 6(c)]

x and y are two variables with $\Sigma x = 25.7$, $\Sigma x^2 = 88.31$, $\Sigma y = 14.40$, $\Sigma y^2 = 26.4324$, $\Sigma xy = 46.856$, $n = 15$. Compute the coefficient of correlation between x and y . Does the value change when each value of x and y is doubled?

[2023–24 | Part 6(a)]

Toughness and fibrousness study of asparagus: x = shear force (kg), y = percent fiber dry weight. x : 46, 48, 55, 57, 60, 72, 81, 85, 94, 109; y : 2.18, 2.10, 2.13, 2.28, 2.34, 2.53, 2.28, 2.62, 2.63, 2.50. (i) Calculate the sample correlation coefficient. (ii) If shear force is expressed in pounds, what happens to r ? Why?

[2022–23 | Part 6(a)]

The Turbine Oil Oxidation Test (TOST) and Rotating Bomb Oxidation Test (RBOT) data for 10 oil specimens (x = TOST hours, y = RBOT minutes): x : 4200, 3600, 3750, 3675, 4050, 2770, 4870, 4500, 3450, 2700; y : 370, 340, 375, 310, 350, 200, 400, 375, 285, 225. (i) Calculate the sample correlation coefficient. (ii) If RBOT is also measured in hours, what happens to r ? Why?

[Jun 2025 | Part 6(b)]

Medical research data: x = maternal age (yr), y = baby's birth weight (kg). x : 15, 17, 18, 15, 16, 19, 17, 16, 18, 19; y : 2.3, 3.4, 3.3, 2.6, 2.9, 3.3, 2.9, 2.5, 3.1, 3.6. Compute the coefficient of correlation and describe the nature of relationship between the two variables.

6.2 Linear Regression

[2025–26 | Part 6(b)]

Data on hours studied (x) and French test score (y) for 10 persons: x : 4, 9, 10, 14, 4, 7, 12, 22, 1, 17; y : 31, 58, 65, 73, 37, 44, 60, 91, 21, 84. (i) Find the equation of the least squares regression line. (ii) Predict the average test score for a person who studied 14 hours.

[2024–25 | Part 6(b)]

Data on tank temperature (x) and efficiency ratio (y) for steel specimens in a phosphating tank. (i) Determine the equation of the estimated regression line. (ii) Calculate a point estimate for true average efficiency ratio when tank temperature is 182.

[2023–24 | Part 6(b)]

Experiment on mozzarella cheese behavior with temperature: x = temperature, y = elongation (%) at failure. x : 59, 63, 68, 72, 74, 78, 83; y : 118, 182, 247, 208, 197, 135, 132. (i) Determine the equation of the estimated regression line. (ii) Estimate the elongation at failure when temperature is 70.

[2022–23 | Part 6(b)]

Data on x = rainfall volume (m^3) and y = runoff volume (m^3): x : 5, 12, 14, 17, 23, 30, 40, 47, 55, 67; y : 4, 10, 13, 15, 15, 25, 27, 46, 38, 46. (i) Determine the equation of the estimated regression line. (ii) Estimate the runoff volume when rainfall volume is 70.

[Jun 2025 | Part 6(c)]

Data on fracture strength x (as % of ultimate tensile strength) and attenuation y (neper/cm) in fiberglass-reinforced polyester composites: x : 12, 30, 36, 40, 45, 57, 62, 67, 71, 78; y : 3.3, 3.2, 3.4, 3.0, 2.8, 2.9, 2.7, 2.6, 2.5, 2.6. Find the equation of regression line and estimate the attenuation when fracture strength is 50.